S SELVA GANESH

Software Engineer | Full-Stack Developer | AI & ML Enthusiast

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PROFESSIONAL SUMMARY

Computer Science and Engineering student specializing in Artificial Intelligence and Machine Learning with experience in full-stack web development, machine learning, and IoT. Skilled in building scalable applications using React, Node.js, Python, PostgreSQL, and modern development frameworks. Passionate about solving real-world problems, developing innovative software solutions, and continuously learning emerging technologies.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, C++
Frontend: React.js, HTML5, CSS3, Tailwind CSS
Backend: Node.js, Express.js, FastAPI, Flask, REST APIs, JWT Authentication
Machine Learning: Scikit-learn, Pandas, NumPy
Databases: PostgreSQL, MongoDB
Embedded Systems & IoT: Arduino, Sensors, RTC Modules, Bluetooth (HC-05), Proteus
Desktop Development: Tkinter
Tools: Git, GitHub, Docker, Vercel, VS Code, Arduino IDE

PROJECTS

Smart Medicine Container & Reminder System | *Arduino Uno, RTC, Load Cell (HX711), Bluetooth (HC-05), Proteus*

- Co-invented an IoT medication-adherence device (patent filed) combining scheduled reminders with physical intake verification.
- Used a load cell to detect whether medicine was actually removed from the container, not just whether an alarm was dismissed.
- Designed escalating alerts: local buzzer/LCD reminder, followed by a Bluetooth notification to a caregiver's phone if the dose is missed.
- Simulated and validated the RTC-based alarm circuit and load-cell detection circuit end-to-end in Proteus.

FundBridge — Full-Stack Crowdfunding Platform | *React.js, Node.js, Express.js, PostgreSQL, JWT*

- Developed and deployed a secure full-stack crowdfunding platform enabling campaign creation, donations, and payout management
- Implemented JWT-based authentication and protected REST APIs for campaigns, donations, and payouts.
- Designed a PostgreSQL schema for users, campaigns, donations, and payout verification.
- Built a responsive UI with search, filtering, and pagination; integrated transactional email via the Resend API.

AI-Based Disease Prediction System | *Python, Scikit-learn, Pandas, NumPy*

- Developed a supervised machine learning model to predict diseases from patient-reported symptoms.
- Performed data preprocessing, feature engineering, and model evaluation using multiple classification algorithms.
- Built a desktop interface for generating predictions from user input.

Real-Time Chat Application | Flask, Flask-SocketIO, Render

- Developed and deployed a real-time chat application supporting multi-user messaging.
- Implemented real-time communication using flask_socketio and deployed the application on Render.

Easy JSON — JSON Viewer & Editor | *React.js*

- Developing a React application for viewing and editing deeply nested JSON structures.
- Implementing recursive rendering, search, and JSON export functionality.

EDUCATION

Lovely Professional University
B.Tech, Computer Science & Engineering (Artificial Intelligence & Machine Learning) | CGPA: 8.07 | Expected Graduation: 2028

ACHIEVEMENTS & INTELLECTUAL PROPERTY

- Co-inventor on a filed invention disclosure: "Smart Medicine Container and Reminder System with Load-Based Intake Detection and Remote Alerting.".
- Strong, applied understanding of REST API design and authentication (JWT).
- Designed and developed multiple full-stack applications independently using modern web technologies.

LANGUAGES

English, Tamil, Hindi